

Yiren (Aaron) Wang

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Engineering

Expected Graduation: May 2027

Relevant Courses: Machine Learning, Algorithms and Models, Data Structures, Computer Systems, Electronic Circuits, Embedded Lab, Signal Processing, Virtual Reality

SKILLS

Programming Languages: Python, C/C++, Java, JavaScript/TypeScript, SQL, Swift, Assembly

Frontend: React, Next.js, HTML/CSS, JavaScript/TypeScript

Backend: FastAPI, Express.js, Node.js

ML / AI: TensorFlow, PyTorch, OpenCV

Tools / Platforms: Git, Linux, Docker, Nginx, AWS (EC2, RDS, S3), Elasticsearch

Databases: MySQL, SQLite

EXPERIENCE

BeeBee AI (backed by Camellia VC)

Remote, US

Full Stack Software Engineer Intern

Jun 2025 - Aug 2025

- Co-developed and launched an AI-powered interview preparation platform that transformed job descriptions into personalized prep plans, voice-based mock interviews, and shareable reports, supporting **800+ sessions in the first week**.
- Built a **RAG pipeline** in Python with **Elasticsearch** vector search, supporting document ingestion, embedding storage, and citation-aware retrieval for responses powered by Mistral-7B and InternVL.
- Developed end-to-end full-stack workflows across user-facing features and backend services, integrating React-based interfaces with **FastAPI** for state management, retrieval, and report generation.
- Built and deployed production service infrastructure with **MySQL, AWS, Docker, and Nginx**, maintained **CI/CD** pipelines with GitHub Actions, and supported API and integration testing to improve deployment reliability across a 5-engineer team.

University of Illinois Urbana-Champaign

Champaign, IL

Undergraduate Researcher - Automated Imaging Systems

Feb 2025 - Sep 2025

- Built an automated microscopy imaging pipeline for particle-size analysis, implementing grayscale preprocessing, flat-field illumination correction, image tiling, and batch export of large TIFF/PNG/JPG datasets.
- Used FIJI to enable high-throughput image conversion and illumination correction across microscopy datasets.
- Applied and tuned **StarDist** for instance segmentation and improved accuracy by ~40% by refining preprocessing (flat-field correction, tiling) and tuning key parameters (prob_thresh, nms_thresh); validated ~420 objects per image.
- Ran GPU batch inference on large tiled images, producing masks, centroids, and overlay outputs; optimized the inference workflow for **3× higher throughput** and exported structured CSV/NPY data for downstream analysis.

PROJECTS

RISC-V Operating System

Jan 2026 - Present

C, RISC-V, QEMU, GDB

- Built core components of a teaching operating system for RISC-V, including trap handling, interrupt-driven I/O, dynamic memory support, and kernel-level threading.
- Implemented and debugged device drivers for UART, RTC, and VirtIO-based devices to support console I/O, timer functionality, and entropy input.
- Developed concurrent kernel functionality using locks, thread coordination, and interrupt-safe synchronization mechanisms.
- Used QEMU and GDB-based low-level debugging to diagnose race conditions, device state issues, and memory faults.

Android Native Media Processing Projects

Nov 2025 – Mar 2026

C++, Python, Android NDK, OpenCV

- Implemented media-processing modules across course projects, including FIR/notch audio filtering, FFT/STFT-based spectrogram generation, image-processing pipelines, and video-analysis experiments.
- Prototyped modules in Python and integrated selected components into Android-native C++/NDK workflows, testing outputs across audio, image, and video inputs to verify functionality and performance.